

Paper Validation Report

for DiMag



Correspondence percentages: 22%

Number of experiments found: 15

Experiment 1. Formulation stated: "we manually collected 1282 raw samples from 126 unique scientific articles (Table S1, Supporting Information) with descriptions of synthesis, materials characterizations, and measurements of magnetic hyperthermia and MRI performances in vitro."

Implementation found: ['script.py:run_ml_pipeline()', 'r2_relaxivity_model.py:train_r2_model()']

Missing components: ['Manual data collection from 126 unique scientific articles', 'Extraction and parsing of synthesis and material characterization descriptions', 'In vitro magnetic hyperthermia and MRI measurement protocols', 'Compilation and curation of the 1282-sample dataset (Table S1)']

Correspondence: 15.0%

Experiment 2. Formulation stated: "we substantially expanded the set of characteristics of magnetic nanoparticles by feature engineering" **Implementation found:** ['NOT FOUND']

Missing components: ['Feature engineering pipeline to expand magnetic nanoparticle characteristics', 'Generation of derived, polynomial, or interaction features from base descriptors', "Any code demonstrating the 'substantial expansion' of the characteristic set as stated in the description"]

Correspondence: 0.0%

Experiment 3. Formulation stated: "we calculated and analyzed the Pearson correlation matrix"

Implementation found: ['NOT FOUND'] **Missing components:** ['Pearson correlation matrix calculation', 'Correlation analysis and statistical evaluation', 'Implementation of correlation visualization or reporting']

Correspondence: 0.0%

Experiment 4. Formulation stated: "we chose kNN imputation technique" **Implementation**

found: ['NOT IMPLEMENTED'] **Missing components:** ['kNN imputation technique for handling missing values (completely absent from all provided scripts)']

Correspondence: 0.0%

Experiment 5. Formulation stated: "we first removed the total of 211 samples such that each sample in the remaining dataset had no more than 15% of missing values. After that, we applied the kNN imputation algorithm to fill the gaps and obtain the complete dataset." **Implementation found:**

['NOT IMPLEMENTED'] **Missing components:** ['Removal of 211 samples based on >15% missing value threshold', 'Application of kNN imputation algorithm to fill missing values']

Correspondence: 0.0%

Experiment 6. Formulation stated: "we transformed highly correlated geometrical features into area to volume and max to min length ratios reflecting size and shape anisotropy of nanoparticles."

Implementation found: ['NOT IMPLEMENTED'] **Missing components:** ['Transformation of highly correlated geometrical features into area-to-volume ratios', 'Transformation of highly correlated geometrical features into max-to-min length ratios', 'Feature engineering logic for calculating nanoparticle size and shape anisotropy']

Correspondence: 0.0%

Experiment 7. Formulation stated: "we discarded noninformative features of low variance, i.e., coercivity and remanent magnetization values for prediction of MRI relaxivities." **Implementation**

found: ['NOT FOUND'] **Missing components:** ['Variance-based feature selection step to identify and discard low-variance features', 'Explicit removal of coercivity and remanent magnetization columns from the feature matrix', 'Integration of feature filtering prior to MinMaxScaler normalization in the preprocessing pipeline']

Correspondence: 10.0%

Experiment 8. Formulation stated: "we removed samples with outliers by setting empirical thresholds on feature values while analyzing their distributions." **Implementation found:** ['NOT

FOUND'] **Missing components:** ['Outlier removal by setting empirical thresholds on feature values', 'Distribution analysis of features to determine thresholds', 'Data filtering logic to remove outlier samples based on feature distributions']

Correspondence: 0.0%

Experiment 9. Formulation stated: "we divided it into three, one for each regression problem: prediction of SAR value, r1 or r2 relaxivity containing 460, 355, and 465 samples, respectively"

Implementation found: ['script.py:SAR_regression_pipeline()', 'script.py:r1_relaxivity_regression_pipeline()', 'script.py:r2_relaxivity_regression_pipeline()', 'r2_relaxivity_model.py:r2_relaxivity_regression_pipeline()']

Missing components: ['Explicit

dataset size validation for 460 (SAR), 355 (r1), and 465 (r2) samples', 'Unified experiment script handling the three-way data division in a single workflow (currently fragmented across multiple independent files)'] **Correspondence:** 85.0%

Experiment 10. Formulation stated: "We normalized and log-scaled the data for each problem before splitting to train and test sets in the 80/20 ratio." **Implementation found:**

['script.py:preprocessing_and_split()', 'script.py:model_training_and_evaluation()', 'r2_relaxivity_model.py:preprocessing_and_split()'],

'r2_relaxivity_model.py:model_training_and_evaluation()'] **Missing components:** ['Exact function names and line numbers NOT FOUND in provided metadata (functions/classes fields are empty)', 'Specific discretization bin thresholds for stratified splitting', 'Exact hyperparameter configurations for ExtraTrees, RandomForest, LightGBM, and XGBoost'] **Correspondence:** 95.0%

Experiment 11. Formulation stated: "hyperparameter tuning was done using the model-agnostic Optuna framework" **Implementation found:** ['NOT IMPLEMENTED'] **Missing components:**

['Optuna framework integration for hyperparameter tuning', 'Hyperparameter search space definitions for ExtraTrees, RandomForest, LightGBM, and XGBoost', 'Optuna objective function and study optimization loop', 'Application of Optuna-tuned hyperparameters to model training pipelines']

Correspondence: 0.0%

Experiment 12. Formulation stated: "The best model was selected using the following metrics: mean Q2 and RMSE for tenfold cross-validation and R2 and RMSE for test samples."

Implementation found: ['script.py:model_training_and_evaluation()'],

'r2_relaxivity_model.py:model_training_and_evaluation()'] **Missing components:** ['Explicit calculation of mean Q2 metric for 10-fold cross-validation (only R2, MAE, MSE, RMSE are computed)', 'Automated best model selection logic based on mean Q2, CV RMSE, test R2, and test RMSE', 'Metric aggregation and comparative ranking step across the four candidate models']

Correspondence: 55.00000000000001%

Experiment 13. Formulation stated: "we plotted SHAP (SHapley Additive exPlanations) values for each selected ML model" **Implementation found:** ['NOT FOUND'] **Missing components:**

['SHAP value calculation and explainer initialization for each selected ML model', 'SHAP visualization/plotting logic for model interpretability', 'Integration of the shap Python library and feature attribution extraction'] **Correspondence:** 0.0%

Experiment 14. Formulation stated: "we formed another testing set of samples (Tables S6 and S7, Supporting Information) that have not been used for training and optimizing hyperparameters of the ML models." **Implementation found:** ['script.py:load_and_evaluate_separate_test_set()'],

'r2_relaxivity_model.py:evaluate_on_test_and_validation_sets()'] **Missing components:** ['Explicit loading and referencing of Tables S6 and S7 as the held-out test set', 'Explicit code logic ensuring the test set is completely excluded from hyperparameter optimization loops', 'Direct mapping of specific CSV filenames to the Supporting Information tables'] **Correspondence:** 65.0%

Experiment 15. Formulation stated: "we computed percent of predictions that differed from the true values by less than 50%." **Implementation found:** ['NOT FOUND'] **Missing components:**

['Computation of percentage of predictions differing from true values by less than 50% (provided scripts only implement standard metrics: R2, MAE, MSE, RMSE)', 'Custom threshold-based accuracy evaluation logic for <50% deviation'] **Correspondence:** 10.0%